

StrideAI — AI Adaptive Running Coach

- Executive technical program case study
- Core differentiation: continuous adaptation using performance, recovery, wearable data and real-world conditions.
- Targets: 200K downloads • >4.7/5 rating • >50% 30-day retention • >70% recommendation acceptance

Product & Technical Strategy

- Prioritize the differentiator: wearable sync, training load, recovery/fatigue, confidence-based adaptation, safety rules and explanation.
- Defer race prediction, weekly summaries and advanced progress tracking.
- Architecture principle: ML/deterministic logic makes the core decision; the LLM explains it.

Six-Week Program Plan

- W1: architecture + APIs + schemas
- W2: data ingestion + first end-to-end slice
- W3: training load + recovery + fatigue
- W4: adaptation + safety + explanation
- W5: integration + performance + monitoring
- W6: internal beta + controlled rollout
- Critical milestone: one real workout produces a basic recommendation by end of W2.

Key Technical Decision

- Option A: LLM directly decides the workout — flexible, but harder to test and guarantee.
- Option B: ML + deterministic logic decide; LLM explains — more predictable, testable and auditable.
- Recommendation: Option B for the MVP because adaptive coaching is safety-sensitive and core to product trust.

Major Program Risk & Trade-off

- Assumed Week-3 recommendation quality: 74% vs 80% production target.
- Commercial pressure: delaying launch could jeopardize a major partnership.
- Recommendation: protect the launch through controlled exposure, not unconditional 100% launch.
- Rollout: Internal → Beta → 20% → 40% → 100%, with quality/safety/reliability gates and rollback criteria.

AI Quality & Safety Framework

- Model quality: precision, recall, F1, calibration and error rates.
- Decision quality: expert evaluation + historical backtesting.
- Safety: critical safety violation rate; target zero.
- User outcomes: acceptance, overrides, satisfaction, retention and adherence.
- Every rollout stage requires evidence-based gates.

Executive Recommendation & TPM Value

- Protect the commercial launch through controlled exposure while protecting the quality bar through explicit gates.
- TPM focus: align functions, identify the critical path, quantify trade-offs, manage RAID/dependencies, establish AI governance and give executives clear decisions.
- Portfolio demonstrates strategy → architecture → execution → AI evaluation → risk → launch governance.